

## **-Business Problem Statement**

The COVID-19 pandemic significantly impacted public transportation in New York City. The goal of this analysis is to understand ridership trends across subways, buses, railroads, and other transit services, and evaluate recovery rates post-pandemic. Insights will help MTA optimize operations, plan for future disruptions, and improve commuter services.

## **-Business Processes Overview**

Data Collection: Daily ridership from MetroCard/OMNY swipes, ticket sales, and traffic counters.

Data Management: Data is validated, stored, and regularly updated.

Reporting: Key ridership metrics are reported to management for operational decisions.

Decision Making: Analysis informs resource allocation, scheduling, and planning strategies.

## **-Stakeholders**

| <b>Stakeholder</b>                         | <b>Role / Interest</b>                      |
|--------------------------------------------|---------------------------------------------|
| MTA Operations Team                        | Optimize service schedules and resources    |
| NYC Government                             | Public transportation planning and policies |
| Commuters                                  | Reliable and efficient transit service      |
| Data Analysts / Business Intelligence Team | Analyze ridership trends and recovery rates |

## **-Metrics & KPIs**

### **Metrics:**

Total Daily Ridership (Subway, Bus, Rail)

% of Pre-Pandemic Ridership

Access-A-Ride Trips

Bridge and Tunnel Traffic

### **KPIs:**

Highest Ridership Day

Lowest Ridership Day

Average Ridership

Recovery Rate (% of Pre-Pandemic)

## -Analytical Questions

How did COVID-19 impact ridership across transport modes?

- Which transportation mode recovered fastest?
- What are the peak and lowest ridership days?
- How do weekdays compare to weekends in ridership patterns?
- Are there seasonal trends or anomalies in ridership?

## - Data Sources

| Source                   | Description                                                                           | File Name                                                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MTA Daily Ridership Data | Daily ridership & traffic for all major MTA agencies, including % vs pre-pandemic day | <a href="https://drive.google.com/file/d/1MAGZ1etB7hD9dq8uL26MReKCbiKm0ot-/view?usp=drive_link">https://drive.google.com/file/d/1MAGZ1etB7hD9dq8uL26MReKCbiKm0ot-/view?usp=drive_link</a> |
| Data Dictionary          | Field definitions, descriptions, and caveats                                          | <a href="https://drive.google.com/file/d/1XJBXens-RKCoF32zJKguAmVTvT-AGZb/view?usp=drive_link">https://drive.google.com/file/d/1XJBXens-RKCoF32zJKguAmVTvT-AGZb/view?usp=drive_link</a>   |
| Overview Document        | Background, methodology, statistical notes, release notes                             |  MTA_DailyRidershipData_Over...                                                                        |

## **-Data Dictionary**

| <b>Column</b>                             | <b>Description</b>                    | <b>Type</b> | <b>Notes</b>                       |
|-------------------------------------------|---------------------------------------|-------------|------------------------------------|
| Date                                      | Date of record                        | Date        | Format YYYY-MM-DD                  |
| Subways: Total Estimated Ridership        | Total subway passengers               | Numeric     | Daily total                        |
| Subways: % of Comparable Pre-Pandemic Day | Subway ridership vs pre-COVID         | Numeric (%) | Comparison with 2019 same day type |
| Buses: Total Estimated Ridership          | Total bus passengers                  | Numeric     | Daily total                        |
| LIRR Ridership                            | Long Island Rail Road passengers      | Numeric     | Daily total                        |
| Metro-North Ridership                     | Metro-North passengers                | Numeric     | Daily total                        |
| Access-A-Ride Trips                       | Trips for eligible disabled customers | Numeric     | Scheduled trips                    |
| Bridges and Tunnels Traffic               | Vehicle traffic                       | Numeric     | Daily total                        |

Staten Island Railway  
Ridership

Staten Island rail  
passengers

Numeric

Daily total

## **-Assumptions & Constraints**

### **Assumptions:**

- Ridership data accurately reflects daily passenger counts.
- % of Pre-Pandemic uses 2019 averages for comparison.
- Data is representative of normal commuting patterns.

### **Constraints:**

- Some missing data or estimation errors may exist.
- Data does not include rider demographics.
- Only NYC public transit is included; private or alternative transport is excluded.